A First Course in Abstract Algebra

Section 6: Cyclic Groups

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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6 Cyclic Groups

Exercise 6.1: Quotient and nonnegative remainder

The division algorithm requires $n = mq + r$ with $0 \leq r < m$. Here

$$42 = 9(4) + 6, \qquad 0 \leq 6 < 9.$$

Thus the quotient is 4 and the remainder is 6. The nonnegative-remainder condition determines the quotient even when the dividend is negative.

Exercise 6.2: Quotient and nonnegative remainder

The division algorithm requires $n = mq + r$ with $0 \leq r < m$. Here

$$-42 = 9(-5) + 3, \qquad 0 \leq 3 < 9.$$

Thus the quotient is -5 and the remainder is 3. The nonnegative-remainder condition determines the quotient even when the dividend is negative.

Exercise 6.3: Quotient and nonnegative remainder

The division algorithm requires $n = mq + r$ with $0 \leq r < m$. Here

$$-50 = 8(-7) + 6, \qquad 0 \leq 6 < 8.$$

Thus the quotient is -7 and the remainder is 6. The nonnegative-remainder condition determines the quotient even when the dividend is negative.

Exercise 6.4: Quotient and nonnegative remainder

The division algorithm requires $n = mq + r$ with $0 \leq r < m$. Here

$$50 = 8(6) + 2, \qquad 0 \leq 2 < 8.$$

Thus the quotient is 6 and the remainder is 2. The nonnegative-remainder condition determines the quotient even when the dividend is negative.

Exercise 6.5: Greatest common divisor: Thirty-two and twenty-four

$32 = 1 \cdot 24 + 8$ and $24 = 3 \cdot 8 + 0$, so $\gcd(32, 24) = 8$. At each Euclidean step, the common divisors of the two current numbers are exactly those of the divisor and remainder; hence the final nonzero remainder is the greatest common divisor.

Exercise 6.6: Greatest common divisor: Forty-eight and eighty-eight

$88 = 1 \cdot 48 + 40$, $48 = 1 \cdot 40 + 8$, and $40 = 5 \cdot 8 + 0$. Thus $\gcd(48, 88) = 8$. At each Euclidean step, the common divisors of the two current numbers are exactly those of the divisor and remainder; hence the final nonzero remainder is the greatest common divisor.

Exercise 6.7: Greatest common divisor: Three hundred sixty and four hundred twenty

$420 = 1 \cdot 360 + 60$ and $360 = 6 \cdot 60 + 0$, so $\gcd(360, 420) = 60$. At each Euclidean step, the common divisors of the two current numbers are exactly those of the divisor and remainder; hence the final nonzero remainder is the greatest common divisor.

Exercise 6.8: Generators at order 5

If a generates a cyclic group of order 5, then a^k is a generator exactly when $\gcd(k, 5) = 1$. Indeed its order is $5/\gcd(k, 5)$. The eligible residues are $\{1, 2, 3, 4\}$, so the number of generators is $\varphi(5) = 4$.

Exercise 6.9: Generators at order 8

If a generates a cyclic group of order 8, then a^k is a generator exactly when $\gcd(k, 8) = 1$. Indeed its order is $8/\gcd(k, 8)$. The eligible residues are $\{1, 3, 5, 7\}$, so the number of generators is $\varphi(8) = 4$.

Exercise 6.10: Generators at order 12

If a generates a cyclic group of order 12, then a^k is a generator exactly when $\gcd(k, 12) = 1$. Indeed its order is $12/\gcd(k, 12)$. The eligible residues are $\{1, 5, 7, 11\}$, so the number of generators is $\varphi(12) = 4$.

Exercise 6.11: Generators at order 60

If a generates a cyclic group of order 60, then a^k is a generator exactly when $\gcd(k, 60) = 1$. Indeed its order is $60/\gcd(k, 60)$. The eligible residues are $\{1, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 49, 53, 59\}$, so the number of generators is $\varphi(60) = 16$.

Exercise 6.12: Automorphisms of Z 2

An additive map from $\mathbb{Z}_2$ is determined by the image k of 1: it must send x to $kx \pmod 2$. It is an automorphism exactly when k generates $\mathbb{Z}_2$, equivalently $\gcd(k, 2) = 1$. The choices are $k \in \{1\}$, so there are 1 automorphisms.

Exercise 6.13: Automorphisms of Z 6

An additive map from $\mathbb{Z}_6$ is determined by the image k of 1: it must send x to $kx \pmod 6$. It is an automorphism exactly when k generates $\mathbb{Z}_6$, equivalently $\gcd(k, 6) = 1$. The choices are $k \in \{1, 5\}$, so there are 2 automorphisms.

Exercise 6.14: Automorphisms of Z 8

An additive map from $\mathbb{Z}_8$ is determined by the image k of 1: it must send x to $kx \pmod 8$. It is an automorphism exactly when k generates $\mathbb{Z}_8$, equivalently $\gcd(k, 8) = 1$. The choices are $k \in \{1, 3, 5, 7\}$, so there are 4 automorphisms.

Exercise 6.15: Automorphisms of the integers

An additive map is determined by $\phi(1)$, since $\phi(n) = n\phi(1)$ for every integer, including negative integers. To be onto, this image must generate $\mathbb{Z}$, so it is 1 or -1. Both maps $n \mapsto n$ and $n \mapsto -n$ are automorphisms. Hence there are exactly 2.

Exercise 6.16: Automorphisms of Z twelve

The image of 1 can be any generator, and determines the whole map by $x \mapsto kx \pmod{12}$. The generators are $1, 5, 7, 11$, the residues relatively prime to 12. Each gives an invertible map (indeed each squares to 1 modulo 12), so there are 4 automorphisms.

Exercise 6.17: Twenty-five modulo thirty

The order of a residue k in the additive group $\mathbb{Z}_n$ is $n/\gcd(n, k)$. Here it is $30/5 = 6$. Directly, the generated subgroup is $\{0, 25, 20, 15, 10, 5\}$; the next multiple is zero and no previous positive multiple is zero.

Exercise 6.18: Thirty modulo forty-two

The subgroup has order $42/\gcd(42, 30) = 42/6 = 7$. Its elements, listed in increasing order, are $\{0, 6, 12, 18, 24, 30, 36\}$. They arise because integer multiples of 30 modulo 42 are exactly the multiples of 6 modulo 42.

Exercise 6.19: The powers of i

The powers are $1, i, -1, -i$, then 1 again. Thus the subgroup has exactly 4 elements. In particular $i^4 = 1$ and no smaller positive power equals 1.

Exercise 6.20: A normalized complex number

The generator $(1 + i)/\sqrt{2} = e^{i\pi/4}$ lies on the unit circle. Its kth power is 1 exactly when $k\pi/4 \in 2\pi\mathbb{Z}$, or $8 \mid k$. Hence its subgroup has 8 elements.

Exercise 6.21: The powers of one plus i

The modulus of $1 + i$ is $\sqrt{2}$. If $(1 + i)^m = (1 + i)^n$, taking moduli gives $(\sqrt{2})^m = (\sqrt{2})^n$, which forces $m = n$. Thus all integral powers are distinct and the generated subgroup is infinite (countably infinite). The modulus need not be 1 in the ambient group $\mathbb{C}^*$.

Exercise 6.22: All subgroups of Z 12

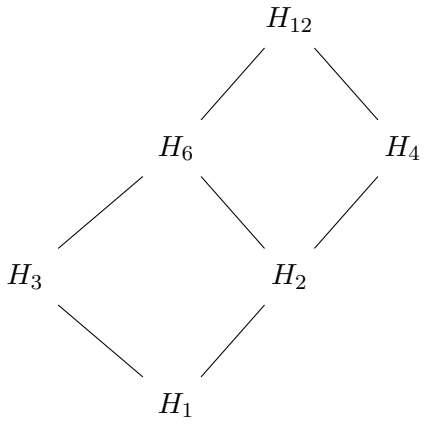
Every subgroup of a cyclic group is cyclic, and there is exactly one subgroup of each order d dividing 12. Label it $H_d = \langle 12/d \rangle$. Its complete element description is

$$H_d = \{0, 12/d, 2(12/d), \ldots, (d-1)(12/d)\}, \qquad |H_d| = d.$$

The complete list is specified by the following table:

d	1	2	3	4	6	12
generator n/d	12	6	4	3	2	1

For $d = 1$, the generator 12 represents 0 modulo 12, so $H_1 = \{0\}$; $H_{12} = \mathbb{Z}_{12}$. Inclusion satisfies $H_d \subseteq H_e$ exactly when $d \mid e$. The diagram joins only immediate inclusions (larger groups above):



Exercise 6.23: All subgroups of Z 36

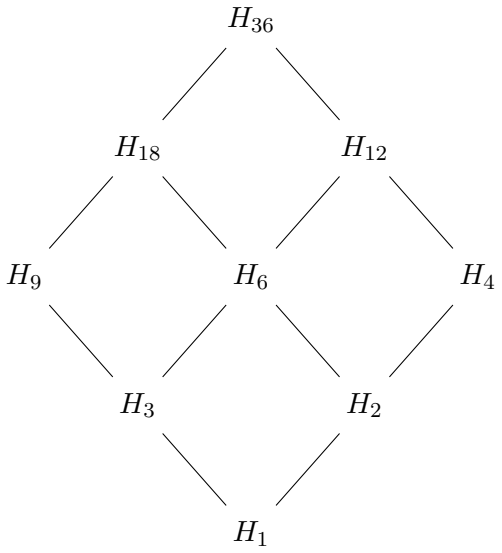
Every subgroup of a cyclic group is cyclic, and there is exactly one subgroup of each order d dividing 36. Label it $H_d = \langle 36/d \rangle$. Its complete element description is

$$H_d = \{0, 36/d, 2(36/d), \ldots, (d-1)(36/d)\}, \qquad |H_d| = d.$$

The complete list is specified by the following table:

d	1	2	3	4	6	9	12	18	36
generator n/d	36	18	12	9	6	4	3	2	1

For $d = 1$, the generator 36 represents 0 modulo 36, so $H_1 = \{0\}$; $H_{36} = \mathbb{Z}_{36}$. Inclusion satisfies $H_d \subseteq H_e$ exactly when $d \mid e$. The diagram joins only immediate inclusions (larger groups above):



Exercise 6.24: All subgroups of Z 8

Every subgroup of a cyclic group is cyclic, and there is exactly one subgroup of each order d dividing 8. Label it $H_d = \langle 8/d \rangle$. Its complete element description is

$$H_d = \{0, 8/d, 2(8/d), \ldots, (d-1)(8/d)\}, \qquad |H_d| = d.$$

The complete list is specified by the following table:

d	1	2	4	8
generator n/d	8	4	2	1

For $d = 1$, the generator 8 represents 0 modulo 8, so $H_1 = \{0\}$; $H_8 = \mathbb{Z}_8$. Inclusion satisfies $H_d \subseteq H_e$ exactly when $d \mid e$. The diagram joins only immediate inclusions (larger groups above):



Exercise 6.25: Subgroup orders in Z 6

The possible orders are exactly the positive divisors of 6:

$$\{1, 2, 3, 6\}.$$

A subgroup of order d actually exists for each listed divisor: it is generated by the residue $6/d$. Conversely the classification of subgroups of a cyclic group excludes all other orders.

Exercise 6.26: Subgroup orders in Z 8

The possible orders are exactly the positive divisors of 8:

$$\{1, 2, 4, 8\}.$$

A subgroup of order d actually exists for each listed divisor: it is generated by the residue $8/d$. Conversely the classification of subgroups of a cyclic group excludes all other orders.

Exercise 6.27: Subgroup orders in Z 12

The possible orders are exactly the positive divisors of 12:

$$\{1, 2, 3, 4, 6, 12\}.$$

A subgroup of order d actually exists for each listed divisor: it is generated by the residue $12/d$. Conversely the classification of subgroups of a cyclic group excludes all other orders.

Exercise 6.28: Subgroup orders in Z 20

The possible orders are exactly the positive divisors of 20:

$$\{1, 2, 4, 5, 10, 20\}.$$

A subgroup of order d actually exists for each listed divisor: it is generated by the residue $20/d$. Conversely the classification of subgroups of a cyclic group excludes all other orders.

Exercise 6.29: Subgroup orders in Z 17

The possible orders are exactly the positive divisors of 17:

$$\{1, 17\}.$$

A subgroup of order d actually exists for each listed divisor: it is generated by the residue $17/d$. Conversely the classification of subgroups of a cyclic group excludes all other orders.

Exercise 6.30: The definition of element order

An element a has order $n \in \mathbb{Z}^+$ if and only if n is the *least positive* integer such that $a^n = e$. Merely requiring $a^n = e$ is not sufficient: if a has order 2, it also satisfies $a^4 = e$, but its order is not 4. If no such positive exponent exists, the order is infinite.

Exercise 6.31: The definition of greatest common divisor

The definition is correct: the greatest common divisor of two positive integers is their largest positive common divisor. Existence is assured because 1 is a common divisor and all positive common divisors are bounded above by the smaller integer. A useful stronger property, proved via the Euclidean algorithm, is that every common divisor divides this greatest common divisor.

Exercise 6.32: Ten statements about cyclic groups

(a) True: powers of a common generator commute.

(b) False: the Klein four-group is abelian but not cyclic.

(c) False: if $q \neq 0$ generated $\mathbb{Q}$ additively, $q/2$ would not be an integer multiple of q.

(d) False: zero in a nontrivial additive cyclic group generates only its trivial subgroup.

(e) True: $\mathbb{Z}_n$ is abelian of order n for every positive n.

(f) False: the Klein four-group has order 4 and is not cyclic.

(g) False: 1 and 9 both generate $\mathbb{Z}_{20}$ but are not prime numbers; relative primality to 20 is the criterion.

(h) False in general: two groups may even have disjoint underlying sets, so their intersection is empty and cannot be a group. An operation shared in an ambient group is not specified here.

(i) True: the intersection contains the shared identity and is closed under the shared operation and inverses.

(j) True: if a generates a cyclic group of order greater than 2, then a^{-1} is a distinct generator; equality would imply $a^2 = e$ and order at most 2. The same reasoning covers infinite order.

Exercise 6.33: A finite noncyclic group

Use the Klein four-group $V = \{e, a, b, c\}$. Each nonidentity element has order 2, so every cyclic subgroup has at most two elements. None can equal the four-element group, proving that it is not cyclic.

Exercise 6.34: An infinite noncyclic group

The additive group $\mathbb{Q}$ is infinite and not cyclic. The zero element cannot generate it. If a nonzero rational q were a generator, then $q/2$ would have to be nq for some integer n; dividing by q gives $n = 1/2$, impossible.

Exercise 6.35: A cyclic group with one generator

The group $\mathbb{Z}_2$ works: the element 1 generates both residues, whereas 0 generates just itself. Hence it has exactly one generator. The trivial group is another possible example, since its identity generates it.

Exercise 6.36: An infinite cyclic group with four generators

No such group exists. Every infinite cyclic group is isomorphic to $\mathbb{Z}$. The only generators of $\mathbb{Z}$ are $1, -1$, since $k\mathbb{Z} = \mathbb{Z}$ forces $|k| = 1$. Isomorphisms preserve generators, so every infinite cyclic group has exactly two.

Exercise 6.37: A finite cyclic group with four generators

Use $\mathbb{Z}_5$. Its generators are $1, 2, 3, 4$, since each is relatively prime to 5 and therefore has order 5. The identity 0 is not a generator, so there are exactly four.

Exercise 6.38: Primitive roots for n equals 4

The roots of unity are $e^{2\pi ik/4}$, and such a root is primitive exactly when $\gcd(k,4)=1$, since its order is $4/\gcd(k,4)$. Therefore the primitive roots are

$$\{e^{2\pi ik/4} : k \in \{1,3\}\}.$$

These are $i, -i$.

Exercise 6.39: Primitive roots for n equals 6

The roots of unity are $e^{2\pi ik/6}$, and such a root is primitive exactly when $\gcd(k,6)=1$, since its order is $6/\gcd(k,6)$. Therefore the primitive roots are

$$\{e^{2\pi ik/6} : k \in \{1,5\}\}.$$

These are $1/2 + i\sqrt{3}/2$ and $1/2 - i\sqrt{3}/2$.

Exercise 6.40: Primitive roots for n equals 8

The roots of unity are $e^{2\pi ik/8}$, and such a root is primitive exactly when $\gcd(k,8)=1$, since its order is $8/\gcd(k,8)$. Therefore the primitive roots are

$$\{e^{2\pi ik/8} : k \in \{1,3,5,7\}\}.$$

These are $(1+i)/\sqrt{2}$, $(-1+i)/\sqrt{2}$, $(-1-i)/\sqrt{2}$, and $(1-i)/\sqrt{2}$.

Exercise 6.41: Primitive roots for n equals 12

The roots of unity are $e^{2\pi ik/12}$, and such a root is primitive exactly when $\gcd(k,12)=1$, since its order is $12/\gcd(k,12)$. Therefore the primitive roots are

$$\{e^{2\pi ik/12} : k \in \{1,5,7,11\}\}.$$

Their arguments are $\pi/6, 5\pi/6, 7\pi/6, 11\pi/6$; equivalently the roots are $(\pm\sqrt{3} \pm i)/2$ with independent signs.

Exercise 6.42: Synopsis of cyclic implies abelian

Express any two elements as powers of one generator and use addition of their integer exponents to see that their products in the two orders agree.

Exercise 6.43: Synopsis of the cyclic-subgroup theorem

For a nontrivial subgroup of a cyclic group, choose the least positive exponent whose generator power belongs to the subgroup. Divide the exponent of any other subgroup element by this least exponent; closure puts the remainder power in the subgroup, and minimality forces that remainder to be zero. Thus every element of the subgroup is a power of the chosen one.

Exercise 6.44: A homomorphism from a cyclic group is determined by one value

Write an arbitrary $x \in G$ as a^m for an integer m. Preservation of products and inverses gives $\phi(x) = \phi(a^m) = \phi(a)^m$. If $\psi(a) = \phi(a)$, then $\psi(x) = \psi(a)^m = \phi(a)^m = \phi(x)$. This works for all x, including the identity and negative powers. Thus the maps agree everywhere once they agree on a generator.

Exercise 6.45: Integer linear combinations form a subgroup

Let $H = \{nr + ms : n, m \in \mathbb{Z}\}$. It contains $0 = 0r + 0s$. The sum of $nr + ms$ and $n'r + m's$ is $(n+n')r + (m+m')s$, again in H. The negative is $(-n)r + (-m)s$, also in H. Hence H is a subgroup of the additive group $\mathbb{Z}$ (indeed it is $\gcd(r, s)\mathbb{Z}$).

Exercise 6.46: The orders of ab and ba

The elements are conjugate: $ba = a^{-1}(ab)a$. Therefore for every positive integer k,

$$(ba)^k = a^{-1}(ab)^k a,$$

proved by cancellation of the adjacent aa^{-1} factors. Consequently $(ba)^k = e$ if and only if $(ab)^k = e$. Their least positive identity exponents are therefore the same, so ba also has order n. This proves equality, rather than merely showing its order divides n.

Exercise 6.47: Least common multiples via subgroups

(a) The intersection $r\mathbb{Z} \cap s\mathbb{Z}$ is a subgroup of $\mathbb{Z}$ containing $rs > 0$. Its unique positive generator is $\ell = \operatorname{lcm}(r, s)$: its positive members are exactly the positive common multiples, and its least positive member generates it.

(b) $\ell = rs$ exactly when $\gcd(r, s) = 1$.

(c) Put $d = \gcd(r, s)$ and write $r = du, s = dv$ with $\gcd(u, v) = 1$. If M is a common multiple, write $M = duk$. Since $dv \mid M$, we have $v \mid uk$; relatively prime u, v imply $v \mid k$ (multiply a Bezout identity for u, v by k). Thus $duv \mid M$. Conversely duv is a common multiple. Therefore $\ell = duv$, and $d\ell = d^2uv = rs$. Since r, s are positive, this formula also proves the equivalence in part (b).

Exercise 6.48: Finitely many subgroups force a finite group

First, no element can have infinite order. If a did, its infinite cyclic subgroup would contain the infinitely many distinct subgroups $\langle a^m \rangle$, $m = 1, 2, \ldots$, all also subgroups of G. They are distinct because in an infinite cyclic group the exponent sets $m\mathbb{Z}$ are distinct. Thus every cyclic subgroup of G is finite. There are only finitely many subgroups by hypothesis, hence only finitely many cyclic subgroups. Every element lies in the cyclic subgroup it generates, so G is a finite union of finite cyclic subgroups and is finite.

Exercise 6.49: All proper subgroups cyclic does not suffice

The Klein four-group is not cyclic, since every element has order at most 2. Its proper subgroups are $\{e\}$, $\{e, a\}$, $\{e, b\}$, and $\{e, c\}$, all cyclic. Indeed any subgroup containing two different nonidentity elements must also contain their product, the third nonidentity element, and hence must be all of V. Thus this is the required counterexample.

Exercise 6.50: A unique element of order two is central

For arbitrary $x \in G$, compute $(xax^{-1})^2 = xa^2x^{-1} = e$. Also $xax^{-1} \neq e$, because otherwise multiplying by x^{-1} on the left and x on the right would give $a = e$. Therefore this conjugate also has order 2, and uniqueness forces $xax^{-1} = a$. Multiplying on the right by x yields $xa = ax$. Since x was arbitrary, a commutes with every element.

Exercise 6.51: Generators at order pq

A residue modulo pq generates the cyclic group exactly when it is not divisible by p or by q. Among the pq residues $0, \ldots, pq-1$, there are q multiples of p, p multiples of q, and one residue (zero) counted in both lists. Inclusion-exclusion gives

$$pq - q - p + 1 = (p-1)(q-1)$$

generators. Distinctness and primality of p, q ensure that these are exactly the forbidden prime divisors.

Exercise 6.52: Generators at a prime-power order

A residue modulo p^r generates the additive group precisely when it is relatively prime to p^r, equivalently not divisible by p. Among $0, 1, \ldots, p^r - 1$, the multiples of p are $0, p, 2p, \ldots, (p^{r-1} - 1)p$, exactly p^{r-1} residues. Therefore the number of generators is

$$p^r - p^{r-1} = p^{r-1}(p-1).$$

This includes $r = 1$, giving $p - 1$ nonzero generators in $\mathbb{Z}_p$.

Exercise 6.53: A power equation when the exponent divides the order

Let $G = \langle a \rangle$ have order n, and write $n = mq$. Every element has a unique representation a^k with $0 \leq k < n$. The equation $(a^k)^m = e$ holds if and only if $n \mid km$, which, since $n = mq$, is equivalent to $q \mid k$. The eligible exponents are therefore

$$0, q, 2q, \ldots, (m-1)q,$$

exactly m distinct residues. The corresponding m distinct group elements are all the solutions. The argument includes the cases $m = 1$ and $m = n$.

Exercise 6.54: A power equation for a general exponent

Put $d = \gcd(n, m)$ and write $n = dn'$, $m = dm'$ with $\gcd(n', m') = 1$. For $x = a^k$, the equation $x^m = e$ is equivalent to $n \mid mk$, hence to $n' \mid m'k$. Relative primality makes this equivalent to $n' \mid k$. There are exactly d such residues modulo n, namely $k = 0, n', \ldots, (d-1)n'$. Thus there are $\gcd(n, m)$ solutions. In the question's case $1 < m < n$ and $m \nmid n$, this number is strictly less than m (and equals 1 if n, m are relatively prime).

Exercise 6.55: Subgroups at prime order

Every subgroup of $\mathbb{Z}_p$ has order dividing p, by the classification of subgroups of a cyclic group. Since p is prime, the only divisors are 1 and p, yielding respectively the trivial subgroup and the whole group. Hence there is no proper nontrivial subgroup. Equivalently, any nonzero residue k is relatively prime to p and itself generates all of $\mathbb{Z}_p$.

Exercise 6.56: Finding a cyclic subgroup of a least-common-multiple order

Write $H = \langle a \rangle$, $K = \langle b \rangle$, with a, b of orders r, s.

(a) Assume $\gcd(r, s) = 1$. Any element of $H \cap K$ has order dividing both r and s, hence is the identity. Since the group is abelian, $(ab)^{rs} = e$. Conversely if $(ab)^t = e$, then $a^t = b^{-t}$ lies in $H \cap K$, so $a^t = b^t = e$. Thus $r \mid t$ and $s \mid t$, giving $rs \mid t$. Therefore ab has order exactly rs, and $\langle ab \rangle$ is the required subgroup.

(b) Factor r, s into prime powers. For each prime p appearing, take the larger of its two prime-power factors, say p^{γ_p}. If that factor comes from r, choose $c_p = a^{r/p^{\gamma_p}}$; its order is exactly p^{γ_p}, by the order formula for a power. If it comes from s, choose $c_p = b^{s/p^{\gamma_p}}$ instead (ties can use either). All these chosen elements commute because G is abelian. Their orders are pairwise relatively prime, so applying part (a) repeatedly shows that the product $c = \prod_p c_p$ has order $\prod_p p^{\gamma_p} = \operatorname{lcm}(r, s)$. Thus $\langle c \rangle$ is the required cyclic subgroup. If $r = s = 1$, simply take $c = e$.

The product ab itself need not have the least-common-multiple order when r, s overlap; the prime-power selection prevents cancellation of common factors.